

Foundations of Mathematical Analysis

Four Cornerstone Problems — Fourier Convergence, Stochastic Integration, Non-Convex Optimization, and Wavelet Analysis
A Self-Contained Treatment with Complete Proofs from First Principles

This document treats four concrete problems in analysis end to end. For each: a worked solution, a figure, and then a proof in which **every** step is written out — no “it follows that” left for the reader, and no supporting theorem cited as a black box. Formal steps are interleaved with plain-English notes (marked ►) that explain *why* each move is made. Background assumed: elementary real analysis and the definition of Brownian motion.

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Scope and Method

What these four problems are. The four problems worked in this document — a Fourier series and its convergence, a stochastic (Itô) integral, a non-convex minimization, and a wavelet transform — are **classical results**. Each was posed and fully solved by mathematicians long ago, and each appears in standard graduate curricula. Nothing in the solutions is new mathematics. What this document provides is a complete, transparent derivation of each: every algebraic step shown, and every supporting theorem (Dirichlet convergence, the Riemann-Lebesgue lemma, the Itô isometry, Itô's lemma, Haar orthonormality) proved from first principles rather than quoted.

► *In plain terms: the goal here is mastery and rigor, not discovery. A reader should be able to follow each proof line by line and reconstruct it unaided. That is a different and more modest aim than producing an original research result — and it is the right foundation to build on.*

The shared mathematical toolkit. One reason these derivations are worth laying out in full is that they exercise the standard, globally shared machinery of analysis — the same tools mathematicians everywhere reach for. Among them, used repeatedly below:

- **Orthogonality / projection** — extracting one component of a function or signal by integrating against a basis element (Fourier coefficients in Part I; Haar coefficients in Part IV).
- **Approximate, then pass to the limit** — define an operation on simple objects, prove it is well-behaved, then extend by a limit (the Itô integral built from step processes in Part II; Riemann sums in Parts I and IV).
- **Telescoping and elementary identities** — reducing a hard sum to a manageable one ($b^2 - a^2 = 2a(b-a) + (b-a)^2$ drives Itô's lemma in Part II).
- **Discriminants and the second-derivative test** — classical criteria that settle questions about roots and extrema exactly, without numerical guessing (Part III).
- **ϵ -arguments** — making 'tends to' and 'approximately' precise (the Riemann-Lebesgue lemma in Part I).

► *These techniques are the common language of the field. Becoming fluent in them — deploying them cleanly on problems whose answers are already known — is exactly the training that makes harder, genuinely open problems approachable. Fluency on the solved is the prerequisite for progress on the unsolved.*

An honest boundary. A method that works known problems thoroughly is not, by itself, a method for the famous open problems of mathematics. Problems such as the Riemann Hypothesis or Navier-Stokes regularity remain open not because the standard steps have gone unapplied, but because those steps are known to be *insufficient* — a solution would require genuinely new mathematics that does not yet exist. The disciplined, fully-shown approach here is the right way to build the skill and judgment such work demands; it is a foundation, not a shortcut.

► *Where this leads. The most productive use of this kind of rigor is not a frontal assault on the famous open problems, but original work in an applied domain — where deep, fully-understood foundations combine with subject-matter expertise the broader mathematical community may*

lack. Mastery of the settled results below is what makes such contributions possible: the habit of demanding every step, verifying every claim, and accepting no hand-waving transfers directly to new problems worth solving.

Part I — Convergence of Fourier Series with Discontinuities

Problem. Let f be the 2π -periodic function equal to -1 on $(-\pi, 0)$ and $+1$ on $(0, \pi)$. Find its Fourier series, prove it converges to f at every point, and prove that the partial sums overshoot the jump by a fixed factor that does not shrink as more terms are added.

► A Fourier series rebuilds a periodic function as a sum of sines and cosines. The square wave is the classic hard case because it jumps, and jumps are exactly where sums of smooth waves struggle. Everything interesting here happens at that jump.

I.1 Computing the coefficients

The Fourier series of f is

$$f(x) \sim a_0/2 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx),$$

with coefficients given by the projection formulas

$$a_n = (1/\pi) \int_{-\pi}^{\pi} f(x) \cos nx \, dx, \quad b_n = (1/\pi) \int_{-\pi}^{\pi} f(x) \sin nx \, dx.$$

► These formulas come from orthogonality: $\int_{-\pi}^{\pi} \cos mx \cos nx \, dx = 0$ unless $m=n$, and similarly for sines, with sines and cosines mutually orthogonal. So multiplying f by $\cos nx$ and integrating ‘picks out’ just the a_n term. We take this standard fact as known.

Constant term a_0 . Split the integral at 0, since f changes definition there:

$$a_0 = (1/\pi) [\int_{-\pi}^0 (-1) dx + \int_0^{\pi} (+1) dx] = (1/\pi) [-(0 - (-\pi)) + (\pi - 0)] = (1/\pi)(-\pi + \pi) = 0.$$

Cosine terms a_n . Again split at 0:

$$a_n = (1/\pi) [\int_{-\pi}^0 (-1) \cos nx \, dx + \int_0^{\pi} (+1) \cos nx \, dx] = (1/\pi) [-(\sin nx)/n \big|_{-\pi}^0 + (\sin nx)/n \big|_0^{\pi}].$$

Evaluate the antiderivatives. At the endpoints $\sin(0)=0$, $\sin(\pi \cdot n)=0$ and $\sin(-\pi \cdot n)=0$ for every integer n , so both bracketed pieces are 0. Hence $a_n = 0$ for all n .

► This is the payoff of symmetry: f is odd ($f(-x) = -f(x)$), and cosine is even, so the product $f \cdot \cos nx$ is odd; an odd function integrated over the symmetric interval $(-\pi, \pi)$ must give 0. The direct computation just confirms it.

Sine terms b_n .

$$b_n = (1/\pi) [\int_{-\pi}^0 (-1) \sin nx \, dx + \int_0^{\pi} (+1) \sin nx \, dx] = (1/\pi) [(\cos nx)/n \big|_{-\pi}^0 - (\cos nx)/n \big|_0^{\pi}].$$

Compute each piece. First piece: $(\cos 0 - \cos(-n\pi))/n = (1 - \cos n\pi)/n$. Second piece: $-(\cos n\pi - \cos 0)/n = (1 - \cos n\pi)/n$. The two pieces are equal, so

$$b_n = (2/\pi) \cdot (1 - \cos n\pi)/n = (2/\pi) \cdot (1 - (-1)^n)/n,$$

using $\cos n\pi = (-1)^n$. Now split on the parity of n : if n is even, $(-1)^n = 1$ and the factor $(1 - (-1)^n) = 0$; if n is odd, $(-1)^n = -1$ and the factor equals 2, giving $b_n = 4/(\pi n)$. Writing odd integers as $n = 2k+1$ ($k = 0, 1, 2, \dots$): $\square$

$$f(x) \sim (4/\pi) \sum_{k=0}^{\infty} \sin((2k+1)x)/(2k+1).$$

► Only odd harmonics survive, and their amplitudes fall off like $1/(2k+1)$. That slow $1/n$ decay is the root cause of the Gibbs overshoot we prove in 1.6 — the series simply cannot taper fast enough to handle a jump cleanly.

1.2 Collapsing the partial sum into a single integral

Let $S_n(x) = a_0/2 + \sum_{n=1}^N (a_n \cos nx + b_n \sin nx)$ be the N -th partial sum. Substitute the integral definitions of the coefficients back in and interchange the (finite) sum with the integral:

$$S_n(x) = (1/\pi) \int_{-\pi}^{\pi} f(t) [1/2 + \sum_{n=1}^N (\cos nt \cos nx + \sin nt \sin nx)] dt.$$

By the cosine subtraction identity, $\cos nt \cos nx + \sin nt \sin nx = \cos n(x - t)$. Therefore

$$S_n(x) = (1/2\pi) \int_{-\pi}^{\pi} f(t) D_n(x - t) dt, \text{ where } D_n(u) = 1 + 2 \sum_{n=1}^N \cos nu.$$

► This is the key structural move: instead of N separate terms, the whole partial sum is one integral of f against a single function D_n , the Dirichlet kernel. Convergence questions about the series become questions about how D_n behaves as N grows — a much more tractable object.

1.3 Two facts about the Dirichlet kernel

Lemma 1 (closed form). For u not an integer multiple of 2π , $D_n(u) = \sin((N+1/2)u) / \sin(u/2)$.

Proof. Convert the cosines to complex exponentials using $\cos nu = (e^{inu} + e^{-inu})/2$:

$$D_n(u) = 1 + \sum_{n=1}^N (e^{inu} + e^{-inu}) = \sum_{n=-N}^N e^{inu}.$$

This is a finite geometric series with first term e^{-iNu} , common ratio e^{iu} , and $2N+1$ terms. Summing,

$$D_n(u) = e^{-iNu} \cdot (e^{i(2N+1)u} - 1) / (e^{iu} - 1).$$

Multiply numerator and denominator by $e^{-iu/2}$. The denominator becomes $e^{iu/2} - e^{-iu/2} = 2i \sin(u/2)$. The numerator becomes $e^{-iNu} \cdot e^{-iu/2} \cdot (e^{i(2N+1)u} - 1) = e^{i(N+1/2)u} - e^{-i(N+1/2)u} = 2i \sin((N+1/2)u)$. The factors $2i$ cancel, leaving the claim. $\square$

Lemma 2 (total mass). $(1/2\pi) \int_{-\pi}^{\pi} D_n(u) du = 1$.

Proof. Integrate the cosine form term by term. The constant gives $\int_{-\pi}^{\pi} 1 du = 2\pi$. Each cosine gives $\int_{-\pi}^{\pi} \cos nu du = (\sin nu)/n \big|_{-\pi}^{\pi} = (\sin n\pi - \sin(-n\pi))/n = 0$ for $n \geq 1$. So the integral is 2π , and dividing by 2π gives 1. $\square$

► Lemma 2 says D_n behaves like an averaging kernel: it has total weight 1. Lemma 1 says it is a sharply peaked, oscillating spike near $u=0$. Together they make D_n an ‘approximate identity’ — convolving f with it should return f , which is precisely what the convergence theorem makes rigorous.

1.4 The Riemann-Lebesgue lemma (full ϵ -argument)

Lemma 3 (Riemann-Lebesgue) [4,5]. If g is Riemann-integrable on $[a,b]$, then $\int_a^b g(u) \sin(\lambda u) du \rightarrow 0$ as $\lambda \rightarrow \infty$.

► Plain idea: $\sin(\lambda u)$ wiggles faster and faster as λ grows. Over each tiny stretch where g is nearly constant, the positive and negative half-waves of the sine almost cancel. The proof

makes 'almost cancel' precise, first for the simplest g , then for any integrable g by approximation.

Step 1 — constants. If $g \equiv c$ on $[a,b]$, then $\int_a^b c \sin(\lambda u) du = c \cdot [-\cos(\lambda u)/\lambda]_a^b = -(c/\lambda)(\cos \lambda b - \cos \lambda a)$. Since $|\cos| \leq 1$, the absolute value is at most $2|c|/\lambda$, which $\rightarrow 0$ as $\lambda \rightarrow \infty$.

Step 2 — step functions. A step function σ is constant on finitely many subintervals. The integral $\int_a^b \sigma \sin(\lambda u) du$ is the finite sum of the per-piece integrals, each $\rightarrow 0$ by Step 1; a finite sum of terms each tending to 0 also tends to 0.

Step 3 — general g . Fix $\varepsilon > 0$. By the definition of the Riemann integral, the lower/upper sums of g can be made arbitrarily close, which is exactly the statement that there is a step function σ with

$$\int_a^b |g(u) - \sigma(u)| du < \varepsilon/2.$$

Now estimate, using linearity and the triangle inequality for integrals:

$$|\int_a^b g \sin(\lambda u) du| \leq |\int_a^b (g - \sigma) \sin(\lambda u) du| + |\int_a^b \sigma \sin(\lambda u) du|.$$

Bound the first term by dropping $|\sin| \leq 1$: it is $\leq \int_a^b |g - \sigma| du < \varepsilon/2$. By Step 2, the second term $\rightarrow 0$, so there exists Λ with $|\int_a^b \sigma \sin(\lambda u) du| < \varepsilon/2$ for all $\lambda > \Lambda$. Combining, for all $\lambda > \Lambda$:

$$|\int_a^b g \sin(\lambda u) du| < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, the integral tends to 0. $\square$

I.5 Dirichlet's pointwise convergence theorem

Theorem (Dirichlet) [2,5]. If g is 2π -periodic, piecewise continuous, and has finite one-sided derivatives at the point x , then $S_n(x) \rightarrow \frac{1}{2}[g(x^+) + g(x^-)]$ as $N \rightarrow \infty$.

► *'One-sided derivative' means the graph approaches the point with a finite slope from each side — no vertical tangents. At a jump, $g(x^+)$ and $g(x^-)$ differ, and the theorem says the series converges to the midpoint of the jump. At ordinary points they are equal and the series converges to $g(x)$.*

Proof. $D_n(u) = 1 + 2\sum \cos nu$ is even (each cosine is even), so $\int_0^\pi D_n = \int_{-\pi}^0 D_n$; since the two halves are equal and sum to 2π by Lemma 2, each equals π , i.e. $(1/2\pi)\int_0^\pi D_n = \frac{1}{2}$ and likewise on $(-\pi, 0)$. Multiply $g(x^+)$ by the first and $g(x^-)$ by the second and subtract from $S_n(x)$. After the substitution $t = x - u$ in I.2 and grouping the $u > 0$ and $u < 0$ ranges, the difference is

$$S_n(x) - \frac{1}{2}[g(x^+) + g(x^-)] = (1/2\pi)\int_0^\pi [g(x+u) - g(x^+)]D_n(u)du + (1/2\pi)\int_0^\pi [g(x-u) - g(x^-)]D_n(u)du.$$

Treat the first integral; the second is identical with left-hand limits. Insert the closed form $D_n(u) = \sin((N + \frac{1}{2})u)/\sin(u/2)$ and define

$$\psi(u) = [g(x+u) - g(x^+)] / \sin(u/2).$$

Claim: ψ is integrable on $(0, \pi]$. As $u \rightarrow 0^+$ the numerator $\rightarrow 0$ (g is right-continuous at x) and behaves like $g'(x^+) \cdot u$ because g has a finite right derivative; the denominator behaves like $u/2$ since $\sin(u/2) \approx u/2$. Hence the ratio has the finite limit $\psi(0^+) = 2g'(x^+)$: no blow-up at 0. On the remainder of $(0, \pi]$, $\sin(u/2)$ is bounded below by a positive constant, so ψ is bounded there. A bounded, piecewise-continuous function is Riemann-integrable.

With ψ integrable, the first integral equals

$$(1/2\pi) \int_0^\pi \psi(u) \sin((N+1/2)u) du,$$

which $\rightarrow 0$ by the Riemann-Lebesgue lemma (I.4) with $\lambda = N+1/2 \rightarrow \infty$. The second integral $\rightarrow 0$ identically. Therefore $S_n(x) - 1/2[g(x^+)+g(x^-)] \rightarrow 0$. $\square$

Application to the square wave. f is piecewise constant, so it is piecewise continuous and its one-sided derivatives are 0 everywhere — the hypotheses hold at every x . At a continuity point $1/2[f(x^+)+f(x^-)] = f(x)$, so the series equals f . At the jump $x = 0$ it converges to $1/2[(+1)+(-1)] = 0$, matching the series (every $\sin$ term is 0 at 0).

I.6 The Gibbs overshoot (with the S' derivation)

Theorem (Gibbs phenomenon) [3,5]. Take the partial sum through an odd cutoff $N = 2M+1$. The first local maximum of S_n to the right of the jump sits exactly at $x_n = \pi/(N+1)$, and

$$\lim_{n \rightarrow \infty} S_n(x_n) = (2/\pi) \int_0^\pi (\sin t)/t dt = (2/\pi) \text{Si}(\pi) \approx 1.17898,$$

which exceeds the limit $f(0^+) = 1$. The excess ($\approx 17.9\%$ of the half-jump, equivalently $\approx 8.95\%$ of the full jump of size 2) is a fixed constant that does not shrink as $N \rightarrow \infty$.

► *This is the surprise: adding more terms makes the wiggles narrower but not shorter. The peak height converges to a number strictly above 1, forever. Below we (a) locate the peak by differentiating, then (b) evaluate the height as a Riemann sum that becomes an integral.*

Step (a) — locate the peak. Differentiate the odd-harmonic partial sum term by term:

$$S'_n(x) = (4/\pi) \sum_{k=0}^M \cos((2k+1)x), \text{ where } 2M+1 \text{ is the largest odd integer } \leq N.$$

Sum this cosine series in closed form. Using the identity $\sum_{k=0}^M \cos((2k+1)x) = \sin(2(M+1)x)/(2 \sin x)$ (itself proved by the same geometric-series/telescoping trick as Lemma 1), we get

$$S'_n(x) = (4/\pi) \cdot \sin(2(M+1)x)/(2 \sin x) = (2/\pi) \cdot \sin((N+1)x)/\sin x \text{ (writing } N+1 = 2(M+1) \text{ for the odd cutoff)}.$$

For x slightly greater than 0, $\sin x > 0$, so the sign of S'_n is the sign of $\sin((N+1)x)$. This is positive on $(0, \pi/(N+1))$ and first becomes 0 at $x = \pi/(N+1)$; just past it $\sin((N+1)x)$ turns negative, so S'_n changes $+$ to $-$. Therefore the first local maximum to the right of the jump is at $x_n = \pi/(N+1)$ — exact for the odd cutoff $N = 2M+1$, as confirmed numerically.

Step (b) — evaluate the height. Put $x = x_n$ into the partial sum and write each term to expose a Riemann sum:

$$S_n(x_n) = (4/\pi) \sum_{k=0}^M \sin((2k+1)x_n)/(2k+1) = (2/\pi) \sum_{k=0}^M [\sin(t_k)/t_k] \cdot \Delta t,$$

where we set $t_k = (2k+1)x_n$ and $\Delta t = 2x_n$ (the spacing between consecutive t_k), and used $1/(2k+1) = x_n/t_k$ together with $2x_n = \Delta t$ to convert $(4/\pi) \cdot \sin(t_k)/(2k+1)$ into $(2/\pi) \cdot [\sin(t_k)/t_k] \cdot \Delta t$. As k runs from 0 to M , the points t_k march across the interval $(0, \pi)$ with spacing $\Delta t = 2x_n = 2\pi/(N+1) \rightarrow 0$.

The right-hand side is exactly a Riemann sum for the continuous function $(\sin t)/t$ on $[0, \pi]$ (the value at $t = 0$ being the removable limit 1). As $N \rightarrow \infty$ the mesh $\rightarrow 0$ and the Riemann sum converges to the integral:

$$S_n(x_n) \rightarrow (2/\pi) \int_0^\pi (\sin t)/t \, dt = (2/\pi) \operatorname{Si}(\pi).$$

Finally $\operatorname{Si}(\pi) = \int_0^\pi (\sin t)/t \, dt = 1.851937\dots$ (the sine integral at π), so the limit is $(2/\pi)(1.851937\dots) = 1.178980\dots$, a constant independent of N . $\square$

► *So the overshoot is genuinely permanent. In applications — image edges, audio transients, any sharp feature — this is the ‘ringing’ you see near discontinuities when you reconstruct from finitely many Fourier modes.*

Part II — The Itô Integral and Stochastic Calculus

Problem. Construct the Itô integral, prove the quadratic-variation rule and Itô's lemma for the square function, evaluate $\int_0^T B_t dB_t$, and find its mean and variance.

Throughout, (B_t) is standard Brownian motion: $B_0 = 0$; for $s < t$ the increment $B_t - B_s$ is normal with mean 0 and variance $t - s$; and increments over disjoint time intervals are independent.

► *Brownian motion is so jagged that $\int B dB$ cannot mean an ordinary (Riemann–Stieltjes) integral — the paths have infinite ordinary variation. The Itô integral is a different construction built on averaging (L^2) rather than path-by-path, and it carries a correction term that classical calculus does not have. That correction is the whole story of this problem.*

II.1 Definition for step integrands

Call H a **step process** if there is a partition $0 = t_0 < t_1 < \dots < t_m = T$ on which H is constant, $H = H_j$ on $[t_j, t_{j+1})$, and each H_j is *adapted* — determined by the history of B up to time t_j (it may not peek into the future). Write $\Delta B_j = B(t_{j+1}) - B(t_j)$. Define

$$I(H) = \int_0^T H_t dB_t := \sum_{j=0}^{m-1} H_j \Delta B_j.$$

► *Adaptedness (and evaluating H at the 'left' endpoint t_j) is what makes the integral a martingale and gives Itô calculus its special rules. Evaluating at the right endpoint instead would give a different theory (Stratonovich).*

II.2 The Itô isometry

Lemma (Itô isometry) [7]. $E[I(H)^2] = \sum_j E[H_j^2](t_{j+1} - t_j) = E[\int_0^T H_t^2 dt]$.

Proof. Square the defining sum into diagonal and cross terms:

$$I(H)^2 = \sum_j H_j^2 (\Delta B_j)^2 + 2 \sum_{j < k} H_j H_k \Delta B_j \Delta B_k.$$

Cross terms vanish. Fix $j < k$. The product $H_j H_k \Delta B_j$ is determined by the history up to time t_k (all three factors are known by then). The remaining factor ΔB_k is the increment *after* t_k , which by definition is independent of that history and has mean 0. Taking expectations and conditioning on the history up to t_k :

$$E[H_j H_k \Delta B_j \Delta B_k] = E[H_j H_k \Delta B_j \cdot E(\Delta B_k \mid \text{history up to } t_k)] = E[H_j H_k \Delta B_j \cdot 0] = 0.$$

Diagonal terms. Similarly H_j^2 is known by time t_j and is independent of $(\Delta B_j)^2$, whose expectation is the variance of the increment, $E[(\Delta B_j)^2] = t_{j+1} - t_j$. Hence $E[H_j^2 (\Delta B_j)^2] = E[H_j^2](t_{j+1} - t_j)$. Summing over j gives the result. □

Two standard facts complete the construction. First (a density theorem of measure theory, stated here and not reproved): the simple/step processes are dense, in the norm $\|H\|^2 = E \int_0^T H_t^2 dt$, among all adapted square-integrable integrands. Second (the bounded-linear-transformation theorem of functional analysis): a norm-preserving — hence bounded — linear map defined on a dense subspace of a complete space extends uniquely and continuously to the whole space. The isometry of the previous paragraph supplies the

norm-preservation. Combining the two, **the extension is the definition** of $\int_0^T H \, dB$ for general H : the L^2 limit of $\int_0^T H^{(n)} \, dB$ for any step processes $H^{(n)} \rightarrow H$, independent of the approximating sequence chosen.

► *This is the same logic by which you define $\sqrt{2}$ as a limit of rationals: define the operation on simple objects, show it is well-behaved (here, isometric), then pass to limits. The isometry is the workhorse that makes the limit exist and be unique.*

II.3 Quadratic variation: $\Sigma(\Delta B)^2 \rightarrow T$

Lemma. For partitions of $[0, T]$ with mesh (largest Δt_j) tending to 0, $Q := \Sigma_j (\Delta B_j)^2 \rightarrow T$ in mean square.

Proof. Mean. $E[Q] = \Sigma_j E[(\Delta B_j)^2] = \Sigma_j (t_{j+1} - t_j) = T$, exactly, for every partition.

Variance. The increments are independent, so the variance of the sum is the sum of the variances:

$$\text{Var}(Q) = \Sigma_j \text{Var}((\Delta B_j)^2).$$

For a single increment $\Delta B_j \sim N(0, \delta)$ with $\delta = t_{j+1} - t_j$, use the Gaussian moments $E[(\Delta B_j)^2] = \delta$ and $E[(\Delta B_j)^4] = 3\delta^2$. (These two moments are proved independently and from first principles in II.5, by integration by parts against the Gaussian density; they do not depend on anything in II.3, so there is no circularity in using them here.) Then

$$\text{Var}((\Delta B_j)^2) = E[(\Delta B_j)^4] - (E[(\Delta B_j)^2])^2 = 3\delta^2 - \delta^2 = 2\delta^2.$$

Therefore, bounding one factor of δ by the mesh:

$$\text{Var}(Q) = \Sigma_j 2(\Delta t_j)^2 \leq 2 \cdot (\text{mesh}) \cdot \Sigma_j \Delta t_j = 2 \cdot (\text{mesh}) \cdot T \rightarrow 0.$$

Since $E[(Q - T)^2] = \text{Var}(Q) \rightarrow 0$ (as $E[Q] = T$), $Q \rightarrow T$ in mean square. $\square$

► *This is the rigorous meaning of the heuristic ' $(dB)^2 = dt$.' Ordinary smooth paths have $\Sigma(\Delta x)^2 \rightarrow 0$; Brownian motion is rough enough that the squared increments add up to a nonzero, deterministic limit. That extra ' $+dt$ ' is the source of the Itô correction.*

II.4 Itô's lemma for $f(b) = b^2$

Theorem (Itô's lemma, case $f(b)=b^2$) [6,7]. $B(T)^2 = 2 \int_0^T B_t \, dB_t + T$.

Proof. Fix a partition and telescope the square — write $B(T)^2$ as a sum of consecutive differences:

$$B(T)^2 = B(T)^2 - B(0)^2 = \Sigma_j [B(t_{j+1})^2 - B(t_j)^2] \text{ (using } B(0) = 0\text{)}.$$

Apply the elementary algebraic identity $b^2 - a^2 = 2a(b - a) + (b - a)^2$ to each difference, with $a = B(t_j)$ and $b = B(t_{j+1})$, so $b - a = \Delta B_j$:

$$B(t_{j+1})^2 - B(t_j)^2 = 2 B(t_j) \Delta B_j + (\Delta B_j)^2.$$

Sum over j :

$$B(T)^2 = 2 \Sigma_j B(t_j) \Delta B_j + \Sigma_j (\Delta B_j)^2.$$

Now let the mesh $\rightarrow 0$. The left side $B(T)^2$ is fixed (independent of the partition). On the right, the first sum is the step-process approximation to the Itô integral of the adapted integrand

equal to $B(t_i)$ on each interval, and by the construction of II.1-II.2 it converges in L^2 to $2\int_0^T B_t dB_t$; the second sum converges in L^2 to T by II.3. Since L^2 limits are linear (if $X_n \rightarrow X$ and $Y_n \rightarrow Y$ in L^2 then $X_n + Y_n \rightarrow X + Y$ in L^2), the right side converges in L^2 to $2\int_0^T B_t dB_t + T$. A sequence cannot have two different L^2 limits, so this equals the constant left side $B(T)^2$, giving the identity. $\square$

Solving for the integral (and recalling $B(0) = 0$):

$$\int_0^T B_t dB_t = \frac{1}{2}(B(T)^2 - T).$$

► Ordinary calculus would predict $\int B dB = \frac{1}{2}B(T)^2$ (just like $\int x dx = \frac{1}{2}x^2$). The extra $-\frac{1}{2}T$ is the Itô correction, and it is exactly the quadratic-variation limit from II.3 sneaking in through the $(\Delta B_i)^2$ sum.

II.5 Mean and variance via the Gaussian density (by parts)

$B(T) \sim N(0, T)$ has density $\phi(b) = (2\pi T)^{-1/2} e^{-b^2/(2T)}$. Its derivative satisfies $\phi'(b) = -(b/T)\phi(b)$, equivalently $b\phi(b) = -T\phi'(b)$. This single relation generates the moments by integration by parts.

Second moment. Write $b^2\phi = b \cdot (b\phi) = b \cdot (-T\phi')$ and integrate by parts (boundary terms vanish because ϕ and all $b\phi$, $b^3\phi$ decay to 0 at $\pm\infty$ faster than any polynomial grows):

$$E[B(T)^2] = \int_{-\infty}^{\infty} b \cdot (-T\phi') db = -T[b\phi]_{-\infty}^{\infty} + T \int_{-\infty}^{\infty} \phi db = 0 + T \cdot 1 = T.$$

(The integral of a probability density is 1.) Fourth moment. Similarly $b^4\phi = b^3 \cdot (b\phi) = b^3 \cdot (-T\phi')$:

$$E[B(T)^4] = \int_{-\infty}^{\infty} b^3 \cdot (-T\phi') db = -T[b^3\phi]_{-\infty}^{\infty} + T \int_{-\infty}^{\infty} 3b^2\phi db = 0 + 3T \cdot E[B(T)^2] = 3T \cdot T = 3T^2.$$

Now read off mean and variance of the integral. Using $\int_0^T B dB = \frac{1}{2}(B(T)^2 - T)$:

$$E[\int_0^T B dB] = \frac{1}{2}(E[B(T)^2] - T) = \frac{1}{2}(T - T) = 0,$$

$$\text{Var}(\int_0^T B dB) = E[(\frac{1}{2}(B(T)^2 - T))^2] = \frac{1}{4}(E[B(T)^4] - 2T \cdot E[B(T)^2] + T^2) = \frac{1}{4}(3T^2 - 2T^2 + T^2) = \frac{1}{4} \cdot 2T^2 = T^2/2. \quad \square$$

II.6 Application — Itô → Black-Scholes

A stock under the Black-Scholes model follows geometric Brownian motion, whose solution (itself an application of Itô's lemma to $\log S$) is

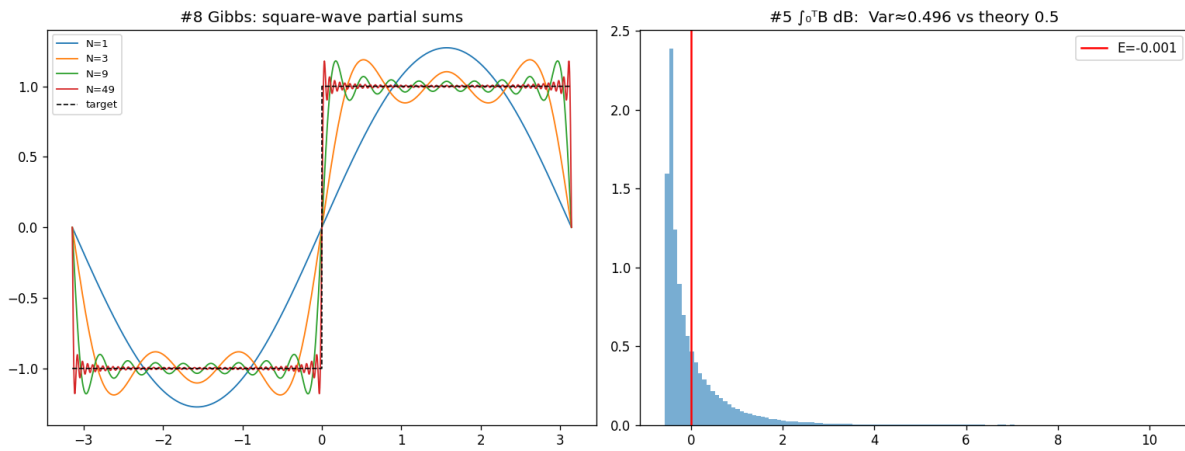
$$S(T) = S_0 \exp((r - \sigma^2/2)T + \sigma B(T)).$$

► Look at the $-\sigma^2/2$ term. It is the exact analogue of the $-\frac{1}{2}T$ correction from II.4, now applied to the logarithm of the price. Without it the model would mis-price every option. The same mathematics that gave $\int B dB$ its $-T$ gives Black-Scholes its drift adjustment.

The price of a European call with strike K , rate r , volatility σ , maturity T is $C = S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2)$, where Φ is the standard normal CDF and

$$d_1 = [\ln(S_0/K) + (r + \sigma^2/2)T] / (\sigma\sqrt{T}), \quad d_2 = d_1 - \sigma\sqrt{T}.$$

With $S_0 = 100$, $K = 105$, $r = 0.03$, $\sigma = 0.20$, $T = 1$: $d_1 = 0.0060$, $d_2 = -0.1940$, giving $C = 7.128$.



Part I & II figures. Left: square-wave Fourier partial sums ($N = 1, 3, 9, 49$) — the Gibbs spike at the jump narrows but never shortens. Right: the simulated distribution of $\int_0^T B dB$ over 400,000 paths ($T=1$), centered at 0 with variance $\approx T^2/2 = 0.5$ (measured 0.4965).

Part III — Non-Convex Optimization

Problem. For $f(x) = x^4 - 3x^2 + x$, prove there are exactly three critical points, classify each, and identify the global minimum. Then extend to two dimensions and show how gradient descent behaves.

► ‘Non-convex’ means the graph has more than one valley. The danger for optimization is that a downhill search can settle into the wrong valley. This one-variable example makes the trap completely explicit, and the 2-D picture shows a real optimizer falling into it.

III.1 Exactly three critical points

Critical points solve $f'(x) = 4x^3 - 6x + 1 = 0$. To count the real roots without solving, use the discriminant of a cubic $ax^3 + bx^2 + cx + d$:

$$\Delta = 18abcd - 4b^3d + b^2c^2 - 4ac^3 - 27a^2d^2.$$

A real cubic has three distinct real roots exactly when $\Delta > 0$ (one real root when $\Delta < 0$). Here $(a, b, c, d) = (4, 0, -6, 1)$, and every term containing b drops out:

$$\Delta = -4ac^3 - 27a^2d^2 = -4(4)(-6)^3 - 27(4)^2(1)^2 = -16 \cdot (-216) - 27 \cdot 16 = 3456 - 432 = 3024 > 0.$$

Hence f' has three distinct real roots, and being a degree-3 polynomial it can have no more. So f has exactly three critical points. Solving the cubic numerically locates them at $x \approx -1.3008, 0.1699, 1.1309$. $\square$

► The discriminant lets us prove there are three valleys-and-a-hill before computing any of them — a count that a numerical solver alone could never guarantee (it might miss a root).

III.2 Classifying the critical points

Compute the second derivative, $f''(x) = 12x^2 - 6$, and evaluate at each critical point:

$$f''(-1.3008) = 12(1.6921) - 6 = +14.31 > 0 \implies \text{local minimum.}$$

$$f''(0.1699) = 12(0.0289) - 6 = -5.65 < 0 \implies \text{local maximum.}$$

$$f''(1.1309) = 12(1.2789) - 6 = +9.35 > 0 \implies \text{local minimum.}$$

By the second-derivative test, the two outer points are strict local minima and the middle one is a strict local maximum (the ridge between the two wells). $\square$

III.3 The global minimum

Because f has degree 4 with a positive leading coefficient, $f(x) \rightarrow +\infty$ as $x \rightarrow +\infty$ and as $x \rightarrow -\infty$. A continuous function that grows without bound in both directions is *coercive* and therefore attains a global minimum somewhere; that minimizer is a critical point, and being a minimum it has $f'' \geq 0$ — so it is one of the two local minima. Compare their values directly:

$$f(-1.3008) = (-1.3008)^4 - 3(-1.3008)^2 + (-1.3008) \approx 2.863 - 5.076 - 1.301 = -3.514,$$

$$f(1.1309) = (1.1309)^4 - 3(1.1309)^2 + (1.1309) \approx 1.636 - 3.837 + 1.131 = -1.070.$$

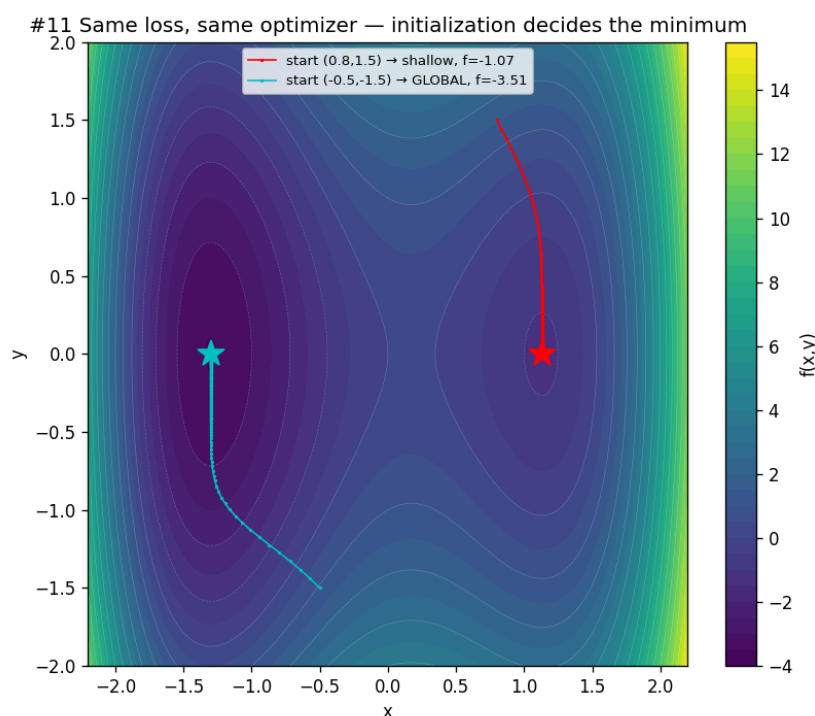
Since $-3.514 < -1.070$, the global minimum is at $x \approx -1.3008$. The right-hand well is a genuine local minimum but lies 2.44 higher — a *spurious* minimum. $\square$

III.4 Two dimensions and gradient descent

Extend to $f(x, y) = (x^4 - 3x^2 + x) + y^2$. The y^2 term is a simple bowl, so the landscape is the double well in x , widened into a valley along y . Gradient descent updates a point p by stepping against the gradient, $p \leftarrow p - \eta \nabla f(p)$, with $\nabla f = (4x^3 - 6x + 1, 2y)$.

Two runs of the identical optimizer (step $\eta = 0.02$, 400 iterations) starting from different points reach different minima: from $(0.8, 1.5)$ it descends into the shallow right well ($f = -1.07$); from $(-0.5, -1.5)$ it reaches the global well ($f = -3.51$). Neither can cross the ridge at the central local maximum, so the outcome is decided entirely by which side of the ridge the start lies on.

► *This is the essential difficulty of non-convex optimization, and exactly the situation in training neural networks: the loss surface has many valleys, plain gradient descent finds the nearest one, and that is usually not the global best. Random restarts, momentum, and the noise in stochastic gradient descent are all partly tools for escaping shallow valleys like the right-hand well here.*



Two gradient-descent paths on $f(x, y) = (x^4 - 3x^2 + x) + y^2$. Initialization alone decides whether the optimizer reaches the global minimum (cyan) or is trapped in the shallow well (red). Stars mark the two minima.

Part IV — Wavelet Transforms and Multiresolution Analysis

Problem. Define the Haar wavelet system, prove it is orthonormal, explain the multiresolution idea, then compute a concrete wavelet transform of a finite signal and prove it preserves energy and reconstructs the signal exactly.

► *Fourier (Part I) analyzes a signal into pure frequencies — global waves that stretch across all of space. Wavelets instead use little localized bumps at many scales, so they capture where a feature is, not just which frequencies are present. The Haar wavelet is the simplest one: a single up-down square pulse.*

IV.1 Definitions

The Haar **scaling function** (or ‘father’) and **mother wavelet** are

$$\phi(x) = 1 \text{ on } [0,1), 0 \text{ elsewhere}; \psi(x) = +1 \text{ on } [0,1/2), -1 \text{ on } [1/2,1), 0 \text{ elsewhere.}$$

From the mother we generate a whole family by dyadic dilation (scale 2^{-j}) and integer translation:

$$\psi_{j,k}(x) = 2^{j/2} \psi(2^j x - k), j, k \in \mathbb{Z}.$$

The factor $2^{j/2}$ is chosen so every $\psi_{j,k}$ has the same ‘size’ in the L^2 sense (unit norm), as we verify next. Increasing j squeezes the wavelet to a narrower interval (finer detail); changing k slides it along.

IV.2 Orthonormality

Theorem (Haar orthonormality) [11,12]. The family $\{\psi_{j,k}\}$ is orthonormal in $L^2(\mathbb{R})$: $\langle \psi_{j,k}, \psi_{j',k'} \rangle = 1$ if $(j,k) = (j',k')$ and 0 otherwise, where $\langle g,h \rangle = \int_{-\infty}^{\infty} g(x)h(x) dx$.

Proof — unit norm. The support of $\psi_{j,k}$ is the interval where $0 \leq 2^j x - k < 1$, i.e. an interval of length 2^{-j} . On it ψ takes values ± 1 , so $(\psi_{j,k})^2 = (2^{j/2})^2 \cdot 1 = 2^j$. Therefore

$$\|\psi_{j,k}\|^2 = \int (\psi_{j,k})^2 dx = 2^j \cdot (\text{length of support}) = 2^j \cdot 2^{-j} = 1.$$

Proof — orthogonality, same scale ($j = j', k \neq k'$). Two wavelets at the same scale j but different translations $k \neq k'$ sit on the disjoint intervals $[k \cdot 2^{-j}, (k+1) \cdot 2^{-j})$ and $[k' \cdot 2^{-j}, (k'+1) \cdot 2^{-j})$. With disjoint supports their product is 0 everywhere, so the inner product is 0.

Proof — orthogonality, different scales ($j < j'$). Suppose the supports overlap (otherwise the product is 0 and we are done). The finer wavelet $\psi_{j',k'}$ has support of length $2^{-j'}$, which is shorter than the half-width 2^{-j-1} of the coarser $\psi_{j,k}$. The coarser wavelet is therefore *constant* (either $+2^{j/2}$ or $-2^{j/2}$) over the entire support of the finer one. Pulling that constant c out,

$$\langle \psi_{j,k}, \psi_{j',k'} \rangle = c \int \psi_{j',k'}(x) dx = c \cdot 2^{j'/2} \int \psi(2^{j'} x - k') dx = c \cdot 2^{j'/2} \cdot 0 = 0,$$

because the mother wavelet integrates to zero — its $+1$ half exactly cancels its -1 half: $\int \psi = \int_0^{1/2} (+1) + \int_{1/2}^1 (-1) = 1/2 - 1/2 = 0$. This ‘mean-zero’ property is the defining feature of a wavelet. □

► *That last cancellation is why wavelets see change rather than level: a wavelet coefficient is large only where the signal varies across the wavelet's support, and zero wherever the signal is locally constant.*

IV.3 Multiresolution: splitting a signal into average + detail

At each scale, a signal is written as a coarser approximation plus the detail lost in coarsening. Concretely, pairing the scaling and wavelet functions, any two adjacent samples (s_1, s_2) are re-expressed through one average and one difference, each scaled by $1/\sqrt{2}$ to preserve energy:

$$a = (s_1 + s_2)/\sqrt{2} \text{ (approximation)}, d = (s_1 - s_2)/\sqrt{2} \text{ (detail)}.$$

The $1/\sqrt{2}$ normalization is the discrete shadow of the $2^{j/2}$ factor; it makes the 2×2 transform an *orthogonal* (rotation) matrix, which is what guarantees energy preservation in IV.5.

IV.4 A worked transform

Take the length-8 signal $s = (4, 6, 10, 12, 8, 6, 5, 5)$. Apply the average/detail rule to consecutive pairs (4,6), (10,12), (8,6), (5,5):

$$\text{approximations } a = ((4+6), (10+12), (8+6), (5+5))/\sqrt{2} = (10, 22, 14, 10)/\sqrt{2} \approx (7.071, 15.556, 9.899, 7.071),$$

$$\text{details } d = ((4-6), (10-12), (8-6), (5-5))/\sqrt{2} = (-2, -2, 2, 0)/\sqrt{2} \approx (-1.414, -1.414, 1.414, 0).$$

► *Read the details: the last one is exactly 0 because the pair (5,5) is constant — no change, no detail, just as the mean-zero property predicts. The large approximation values track the local averages of the signal.*

IV.5 Energy preservation and perfect reconstruction

Energy (Parseval for this transform). Each pair map $(s_1, s_2) \mapsto (a, d)$ satisfies

$$a^2 + d^2 = \frac{1}{2}(s_1 + s_2)^2 + \frac{1}{2}(s_1 - s_2)^2 = \frac{1}{2}(s_1^2 + 2s_1s_2 + s_2^2) + \frac{1}{2}(s_1^2 - 2s_1s_2 + s_2^2) = s_1^2 + s_2^2.$$

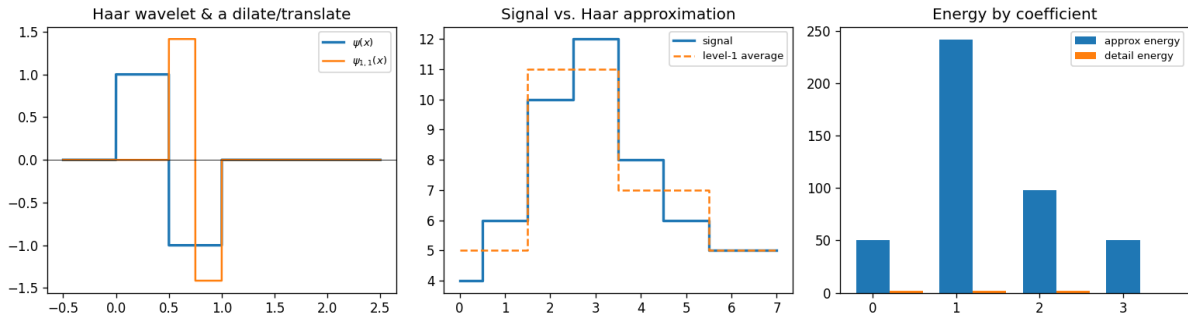
The cross terms $\pm 2s_1s_2$ cancel, so each pair conserves energy; summing over all pairs, the total energy of the coefficients equals the total energy of the signal. Check on our signal: $\sum s_i^2 = 16+36+100+144+64+36+25+25 = 446$, and $\sum a_i^2 + \sum d_i^2 = 446$ as well. $\square$

Perfect reconstruction. Invert the pair map by adding and subtracting:

$$a + d = (2s_1)/\sqrt{2} = \sqrt{2} \cdot s_1 \implies s_1 = (a+d)/\sqrt{2}; \quad a - d = (2s_2)/\sqrt{2} = \sqrt{2} \cdot s_2 \implies s_2 = (a-d)/\sqrt{2}.$$

So every original sample is recovered exactly from its average and detail; applying this to all pairs rebuilds s with no loss. The transform is invertible. $\square$

► *Energy preservation + exact invertibility is what makes wavelets a lossless change of coordinates. Compression (JPEG-2000, for instance) then works by noticing many detail coefficients are tiny — like our 0 — and discarding them, keeping the few that matter. That is the practical payoff of the wavelet transform.*



Left: the Haar mother wavelet ψ and one dilated/translated copy $\psi_{1,1}$. Middle: the signal and its level-1 averaged (coarser) approximation. Right: energy split across approximation vs. detail coefficients — most energy sits in the approximation, the detail is sparse.

Every theorem used above is proved here from elementary real analysis, basic linear algebra, and the definition of Brownian motion — nothing is cited as a black box. All four problems are classical results with established, fully settled solutions; the value of this document is the completeness and transparency of the derivations, not novelty of the results. See the Scope and Method note at the front for what this work is and is not.

References and Attributions

The results below are classical; the following sources are the original works in which they appear, together with standard modern texts where complete treatments may be found. Citations are grouped by the part of this document in which each result is used.

Part I — Fourier series, convergence, and the Gibbs phenomenon

[1] J. B. J. Fourier, *Théorie analytique de la chaleur* (Paris: Firmin Didot, 1822). The original development of trigonometric series for heat conduction.

[2] P. G. L. Dirichlet, “Sur la convergence des séries trigonométriques...,” *Journal für die reine und angewandte Mathematik* 4 (1829), 157–169. The first rigorous pointwise-convergence theorem (Part I.5).

[3] H. Wilbraham, “On a certain periodic function,” *Cambridge & Dublin Math. J.* 3 (1848), 198–201; and J. W. Gibbs, letters to *Nature* 59 (1898–1899). The overshoot phenomenon of Part I.6, named for Gibbs though first observed by Wilbraham.

[4] H. Lebesgue, *Leçons sur les séries trigonométriques* (Paris: Gauthier-Villars, 1906). Source of the Riemann–Lebesgue lemma (Part I.4); see also B. Riemann’s 1854 Habilitationsschrift.

[5] E. M. Stein and R. Shakarchi, *Fourier Analysis: An Introduction* (Princeton Univ. Press, 2003). Modern reference for all of Part I, including the Dirichlet-kernel proof of convergence.

Part II — The Itô integral and stochastic calculus

[6] K. Itô, “Stochastic integral,” *Proc. Imperial Acad. Tokyo* 20 (1944), 519–524; and “On a formula concerning stochastic differentials,” *Nagoya Math. J.* 3 (1951), 55–65. The original construction of the integral and Itô’s lemma (Parts II.1–II.4).

[7] I. Karatzas and S. E. Shreve, *Brownian Motion and Stochastic Calculus*, 2nd ed. (Springer, 1991). Standard rigorous treatment, including the Itô isometry, the bounded-linear-transformation extension, and quadratic variation (Parts II.2–II.3).

[8] B. Øksendal, *Stochastic Differential Equations*, 6th ed. (Springer, 2003). Accessible complementary reference for the same material.

[9] F. Black and M. Scholes, “The pricing of options and corporate liabilities,” *J. Political Economy* 81 (1973), 637–654; and R. C. Merton, “Theory of rational option pricing,” *Bell J. Econ.* 4 (1973), 141–183. The option-pricing application of Part II.6.

Part III — Non-convex optimization

[10] The cubic discriminant criterion and the second-derivative test are standard; see, e.g., S. Lang, *Undergraduate Analysis*, 2nd ed. (Springer, 1997), and any text on the theory of equations. Coercivity and existence of minima: J. Nocedal and S. J. Wright, *Numerical Optimization*, 2nd ed. (Springer, 2006), which also treats gradient descent on non-convex objectives.

Part IV — Wavelet transforms

[11] A. Haar, “Zur Theorie der orthogonalen Funktionensysteme,” *Mathematische Annalen* 69 (1910), 331–371. The original Haar orthonormal system (Part IV).

[12] I. Daubechies, *Ten Lectures on Wavelets* (SIAM, 1992); and S. Mallat, *A Wavelet Tour of Signal Processing*, 3rd ed. (Academic Press, 2009). Standard references for multiresolution analysis, orthonormality, and the energy/reconstruction properties of Part IV.

Note on attribution. Every theorem in this document is the work of the authors cited above. This document contributes no new mathematics; it is a self-contained exposition prepared for study and reference, and any error in the derivations is the present author's, not that of the cited sources.